

# Testing the Dorabella Cipher

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No simple substitution key turns the Dorabella into English or German: across about 30 fair tests, the cipher scores at chance level, while known English run through the same tests is recovered every time.

## Results at a glance

The Dorabella beats plain shuffles of its own symbols, but never beats shuffles that keep its adjacent-pair habits, and never finds more real words than chance. Every one of 41 known-English controls scored far above it.

| Idea tested                                                                                                                    | Script     | Real Dorabella | Scrambled copies (mean / best)          | p              | Verdict                  |
|--------------------------------------------------------------------------------------------------------------------------------|------------|----------------|-----------------------------------------|----------------|--------------------------|
| Any one-symbol-one-letter key, 160 reading configs, vs plain shuffles                                                          | marathon   | z 5.28         | 2.24 / 5.22 (400)                       | 0.0025         | Order is not random      |
| Same, vs pair-preserving shuffles                                                                                              | marathon   | z 5.28         | 4.33 / 6.66 (400)                       | 0.085          | Explained by pair habits |
| Words in the best decrypt                                                                                                      | marathon   | 11 words       | p over 400 copies                       | 0.39           | Chance                   |
| Known English through the same pipeline                                                                                        | marathon   | z 5.28         | 41 controls: median z 14.4, lowest 8.97 | 41 of 41 above | Not English-like         |
| Key learned on 2 lines, scored on the 3rd, vs pair-preserving                                                                  | marathon   | -13.74         | -14.01 / -13.29 (100)                   | 0.27           | Chance                   |
| Deeper search (simulated annealing), vs plain shuffles                                                                         | marathon   | -10.84         | -11.38 / -11.10 (40)                    | 0.024          | Order is not random      |
| Loop count picks the letter group (1 loop = A-H etc.), direction the letter: all 24,576 keys x 160 configs, vs pair-preserving | loopgroups | -15.06         | -15.14 / -14.54 (200)                   | 0.30           | Chance                   |

| Idea tested                                                                                     | Script           | Real Dorabella        | Scrambled copies<br>(mean / best) | p    | Verdict |
|-------------------------------------------------------------------------------------------------|------------------|-----------------------|-----------------------------------|------|---------|
| Keypad-style keys: 287 triplet and affine layouts x 160 configs, vs plain shuffles              | tests (Part C)   | -16.27                | -16.10 / -15.25 (300)             | 0.72 | Chance  |
| All 10.3 million sets-of-three keys x 160 configs (standard grouping)                           | allkeys          | -13.84                | -13.59 / -12.77 (50)              | 0.84 | Chance  |
| Same, alternative grouping (ABC DEF GHI KLM NOP QRS TUW XYZ)                                    | allkeys          | -14.05                | -14.36 / -13.64 (50)              | 0.14 | Chance  |
| Keyword or phrase ordering the sets (261 phrase variants)                                       | phrase / keyword | best p 0.003 (POWICK) | 2 to 3 such hits expected by luck | —    | Chance  |
| TOMORROW as a Nihilist-style shift key                                                          | one-off          | IC 0.0569             | 0.0567 (2,000)                    | 0.44 | Chance  |
| Rare symbols are digits (Courage cards)                                                         | numbers          | -10.80                | -10.99 / -10.56 (100)             | 0.16 | Chance  |
| Rechecked transcription, 10 uncertain symbols solved in the search, English, vs pair-preserving | v2               | -10.88                | -10.93 / -10.41 (200)             | 0.33 | Chance  |
| Same, German (Fontane and two other 1890s novels)                                               | v2               | -10.98                | -11.05 / -10.72 (200)             | 0.30 | Chance  |

Scores are average log-probabilities of four-letter sequences; higher is more language-like. Real English of this length scores about -9.2, real German about -9.4. A p-value can never go below  $1/(\text{copies} + 1)$ .

## The cipher as tested

Each of the 87 symbols is written as a two-digit code: loop count (1–3) then direction (1–8), with direction 1 = opening facing up. The common convention's direction 1 (opening to the right) is direction 3 here. Elgar's standard key, applied to these codes, reproduces the published decrypt (BLTACEIARWUNISNF...) at 80 of 87 positions; the misses fall on uncertain symbols.

line, codes

```
1,23 37 24 33 13 25 11 33 16 22 34 28 18 14 28 35 28 28 25 37 37 28 15 12 22
12 15 34 38
2,11 28 11 25 11 33 16 26 33 24 28 28 24 25 15 18 11 28 34 28 25 21 38 18 14
12 16 16 12 34 38
3,24 38 25 21 38 24 14 21 31 15 38 24 28 25 21 18 38 15 33 37 15 38 25 33 15
13 33
```

A careful recheck against a high-resolution image left 10 symbols uncertain. Every doubt is between two neighboring directions, never the loop count.

| Position | Main reading | Alternate |
|----------|--------------|-----------|
| 10       | 22           | 23        |
| 23       | 15           | 14        |
| 24       | 12           | 11        |
| 25       | 22           | 23        |
| 32       | 11           | 18        |
| 34       | 11           | 12        |
| 67       | 14           | 15        |
| 78       | 15           | 14        |
| 85       | 15           | 14        |
| 86       | 13           | 12        |

The marathon, allkeys, numbers, phrase and keyword runs used the first transcription (five uncertain positions: 23, 32, 34, 78, 86; position 85 read as 14). The v2 runs used the rechecked one above, with all 1,024 combinations of the 10 readings open to the search.

## How the tests work

Every test asks one question: does the real Dorabella do better than scrambled copies of itself put through exactly the same search? That corrects for trying millions of keys, which is where most proposed solutions fail.

1. **Score.** A decrypt is scored by how common its four-letter sequences are in period text: 2.5 million letters of 1890s English (Dracula), or 1.4 million of 1890s German.
2. **Search.** A hill climb (and, in the deep test, simulated annealing) looks for the key that maximizes that score, in each reading order: forward, fully reversed, each line reversed, and two alternating (boustrophedon) orders. Uncertain symbols are tried both ways.
3. **Scramble.** The same search is run on scrambled copies of the cipher. Plain shuffles put the 87 symbols in random order. Pair-preserving shuffles keep every adjacent symbol pair exactly as in the Dorabella, so they keep the writer's habits but carry no language.
4. **Compare.** The p-value is the share of copies that did as well as the real cipher. Beating plain shuffles only shows the order is not random; beating pair-preserving shuffles would point to something beyond habit.
5. **Check independently.** Word counts and held-out scoring (learn the key on two lines, score the third) are not what the search maximizes, so a real solution should pass them too.
6. **Prove the method works.** Known English (and German), enciphered with random keys and pushed through the same pipeline, sometimes with 3 deliberately misread symbols, shows what a genuine message looks like.

## Findings by idea

### Known English is easy to find; the Dorabella is not like it

All 41 English controls scored above the Dorabella (median  $z$  14.4 against 5.28). Clean controls decoded at a median 99% of letters and those with 3 misread symbols at 90%; all 41 picked the right reading order. The Dorabella's best decrypt reads:  
 OCTOSANDONAHUSERANBUTRAUSARAY / ILLITHAUSEYEWHSREEROFLOWSEWAY  
 / NIGINEDDSEEMETHEYGLOWSCORDP.

### The non-random order is writing habit

The Dorabella beat all 400 plain shuffles, but 33 of 400 pair-preserving shuffles matched it. Most of its structure is which symbols tend to follow which, not language.

### Loop count as the letter group

The first model tried: one loop for A-H, two for I/J-Q, three for R-Z (in any assignment), with direction picking the letter in any rotation, each group read either way. All 24,576 such keys in all 160 configurations give a best of -15.06, the same as plain shuffles ( $p =$

0.40) and pair-preserving shuffles ( $p = 0.30$ ), with fewer real words than either. The best decrypt reads INXTCTWEKKHEEMEWRENCYOSHUOAKB...

## Elgar's alphabet in sets of three

Elgar's 1920 notebook groups letters in threes that alternate direction (A-B-C up, D-E-F down...). Every arrangement of the 8 sets around the 8 directions, each read up or down, is  $8! \times 2^8 = 10,321,920$  keys; all were scored in all 160 reading configurations. Neither letter grouping beat chance; the standard grouping scored below the average scrambled copy.

## A keyword Dora would know

No 8-letter word in 19,776 words of period vocabulary fits one letter per set. Treated as Victorian keyword alphabets, 64 phrases (DORABELLA, MISS PENNY, MISDORAPENY, WOLVERHAMPTON, ENIGMA, POWICK...) with no-vowel and initials variants gave only chance-level hits. TOMORROW, missing from the notebook's enciphered sentence, fails both as a keyword and as a Nihilist-style shift.

## A second layer

A repeating shift of the Nihilist kind, which Elgar solved in 1896, flattens symbol frequencies (index of coincidence about 0.045). The Dorabella's is 0.058, inside the range of plain English (0.056 and up). Its frequencies look like a direct, one-layer cipher.

## Numbers

The first Courage card shows ten loop symbols, as if mapping digits. Treating up to three of the six rarest symbols as digits changed nothing: best decrypt "qfsd#ildu#me THE BEE...",  $p = 0.16$ , fewer words than the copies. Planted English with a date and a train time was recovered at 99%.

## The rechecked transcription and German

With the 10 uncertain symbols solved inside the search, English and German both land in the middle of the pair-preserving copies ( $p = 0.33$  and  $0.30$ ). Planted English and German notes were recovered at 97% and 98%.

## Open threads

Three small signals survived, none strong enough to call a result.

- **Position 32 read as 18.** Against pair-preserving shuffles, choosing 18 there helps more than chance ( $p = 0.0075$ , 2 of 400). The English and German v2 searches both chose it independently. It is the one reading worth rechecking on the original first; position 67 as 15 is second.
- **Reversed reading orders.** Fully reversed and each-line-reversed beat pair-preserving shuffles modestly ( $p = 0.015$  and  $0.018$ ). About ten such order and reading comparisons were made, so one or two at this level is expected by chance.
- **A pre-registered configuration** (each line reversed, with 32=18, 34=12, 78=14, 86=12) hit the floor on the held-out test against both kinds of copy ( $p = 0.0099$ ). It was chosen from earlier runs on the same cipher, so this is not an independent confirmation.

The largest remaining unknown is the plaintext, not the key. Elgar's own key gives English-like letter frequencies but not English-like order, which fits phonetic spelling, private abbreviations or in-jokes — none of which a standard English scorer can reward. A scorer built for that style is the natural next test.

## Reproducing this

All scripts, logs and the transcription are in the GitHub repository [cjager13/Dorabella-cipher-tests](#). They are plain Python 3 and run on an 8-core laptop; allkeys also needs numpy. Keep `Dorabella_hillclimb.py`, `Dorabella_tests.py` and `english.txt` in the same folder, since the others load them. Every script has a `--selftest` or `planted-text` check; run it first.

| Script                                                            | What it tests                                                                                                  | Time (8-core laptop) |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------|
| <code>Dorabella_marathon.py</code>                                | Main search, both scramble types, known-English controls, held-out, deep search; resumable, runs until stopped | 18.6 h first pass    |
| <code>Dorabella_allkeys.py --all-configs --nulls 50</code>        | All 10,321,920 sets-of-three keys x 160 configs                                                                | 16.8 h               |
| <code>Dorabella_numbers.py</code>                                 | Rare symbols as digits                                                                                         | 1.6 h                |
| <code>Dorabella_v2.py</code> ( <code>--lang de</code> for German) | Rechecked transcription, uncertain symbols solved in the search                                                | 45 min each          |
| <code>Dorabella_verify.py</code>                                  | 160-config search vs 20 identical searches on shuffles                                                         | 20 min               |

| Script                  | What it tests                                                     | Time (8-core laptop) |
|-------------------------|-------------------------------------------------------------------|----------------------|
| Dorabella_phrase.py     | Keyword and phrase alphabets                                      | 30 s                 |
| Dorabella_keyword.py    | 8-letter keywords, one letter per set                             | seconds              |
| Dorabella_loopgroups.py | All 24,576 loop-group keys vs both kinds of shuffle (needs numpy) | about 10 min         |

The German corpus is spelled out without umlauts (ä to AE, ö to OE, ü to UE, ß to SS) and stripped of Project Gutenberg's English license text.

The first day's outputs (27 September 2026) are in the repository's `early_runs` folder, summarized in its README. One of them matters for everything above: planted English with 2, 4 or 8 misread symbols was recovered at only 87%, 48% and 30% by the early solver, so a few transcription errors could hide a message. That is why the transcription was rechecked and its uncertain symbols built into the v2 search.

## Sources

- [Cipher Mysteries: The Dorabella cipher and Elgar's other little ciphertexts](#) — Courage cards, the 1896 Nihilist cipher, the 1920 notebook
- [Cipher Mysteries: Dorabella Cipher](#) — the standard decrypt
- [NumberWorld: The Dorabella Cipher \(Part 3\)](#)
- [Wikipedia: Dorabella Cipher](#)
- [GitHub: cjager13/Dorabella-cipher-tests](#) — scripts, logs and transcription
- Corpora: Bram Stoker, *Dracula* (1897), and Theodor Fontane, *Effi Briest* (1895), plus two other 1890s German novels, all from Project Gutenberg